



Project Title - Seasonal Agriculture Performance Analysis

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AICTE STU ID : (STU683a1310cc13e1748636432)



PROBLEM STATEMENT

- Agriculture performance depends on several factors such as crop type, season, rainfall, temperature, soil conditions, irrigation method, fertilizer usage and market price. It is difficult to identify the factors affecting crop yield, profitability and water efficiency from raw agricultural data. This project aims to analyze agricultural performance and identify useful patterns through data visualization.



Project Description



The project analyzes a seasonal agriculture dataset containing 4,000 farm records and 28 attributes. The analysis focuses on crop performance, seasonal trends, state-wise performance, irrigation methods, yield, production, revenue, cost, profit, water usage and disease/pest risk. Data visualization is used to convert the raw dataset into meaningful insights that can support better agricultural decision-making

WHO ARE THE END USERS?

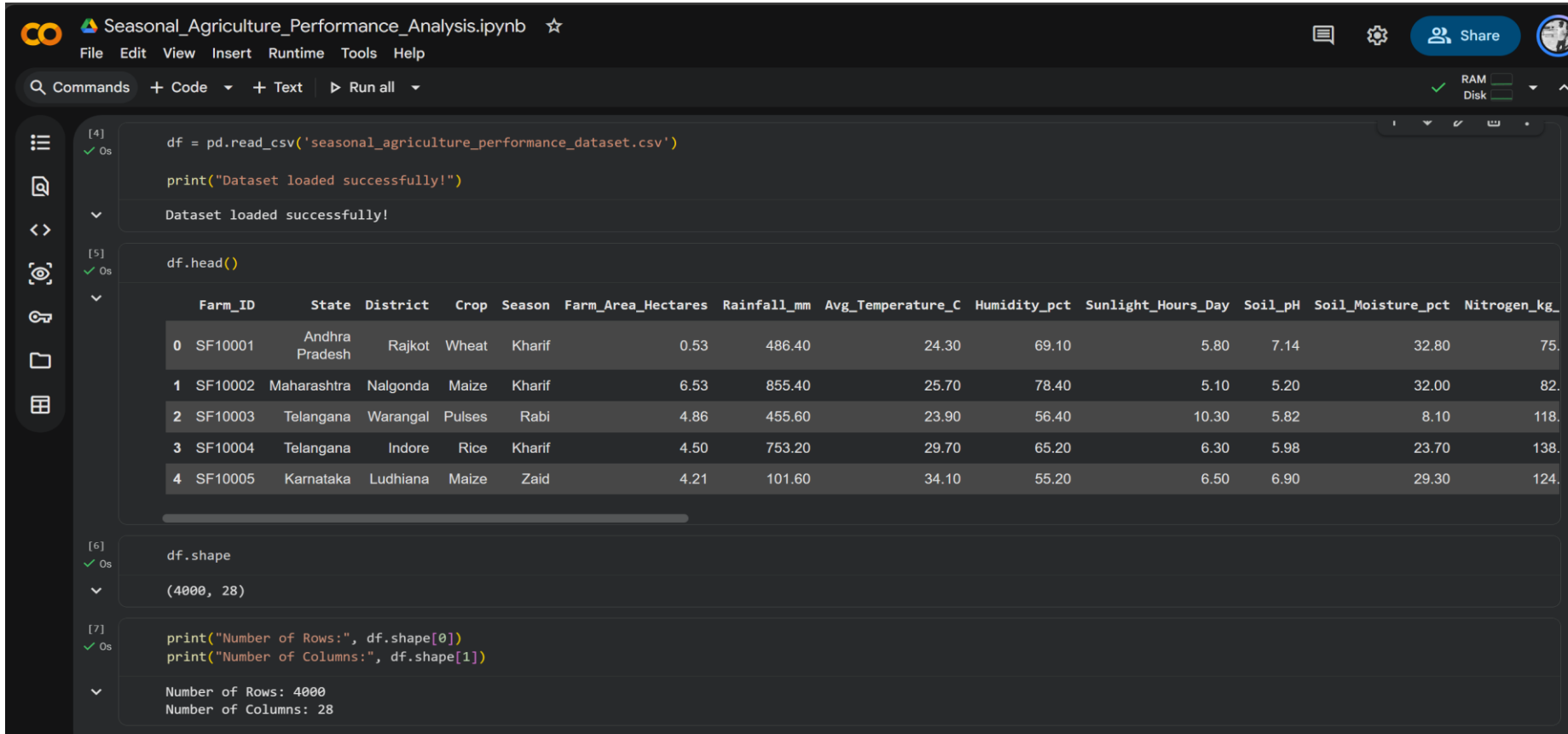
- Farmers
- Agricultural consultants
- Agriculture researchers
- Government agriculture departments
- Irrigation planners
- Agricultural businesses

Technology Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook / Google Colab
- CSV Dataset



RESULTS - Dataset & Data Analysis



The screenshot displays a Jupyter Notebook interface with the following content:

```
[4] df = pd.read_csv('seasonal_agriculture_performance_dataset.csv')
    print("Dataset loaded successfully!")
    Dataset loaded successfully!

[5] df.head()
    Farm_ID  State District Crop Season Farm_Area_Hectares Rainfall_mm Avg_Temperature_C Humidity_pct Sunlight_Hours_Day Soil_pH Soil_Moisture_pct Nitrogen_kg_
0  SF10001  Andhra Pradesh Rajkot Wheat Kharif 0.53 486.40 24.30 69.10 5.80 7.14 32.80 75.
1  SF10002  Maharashtra Nalgonda Maize Kharif 6.53 855.40 25.70 78.40 5.10 5.20 32.00 82.
2  SF10003  Telangana Warangal Pulses Rabi 4.86 455.60 23.90 56.40 10.30 5.82 8.10 118.
3  SF10004  Telangana Indore Rice Kharif 4.50 753.20 29.70 65.20 6.30 5.98 23.70 138.
4  SF10005  Karnataka Ludhiana Maize Zaid 4.21 101.60 34.10 55.20 6.50 6.90 29.30 124.

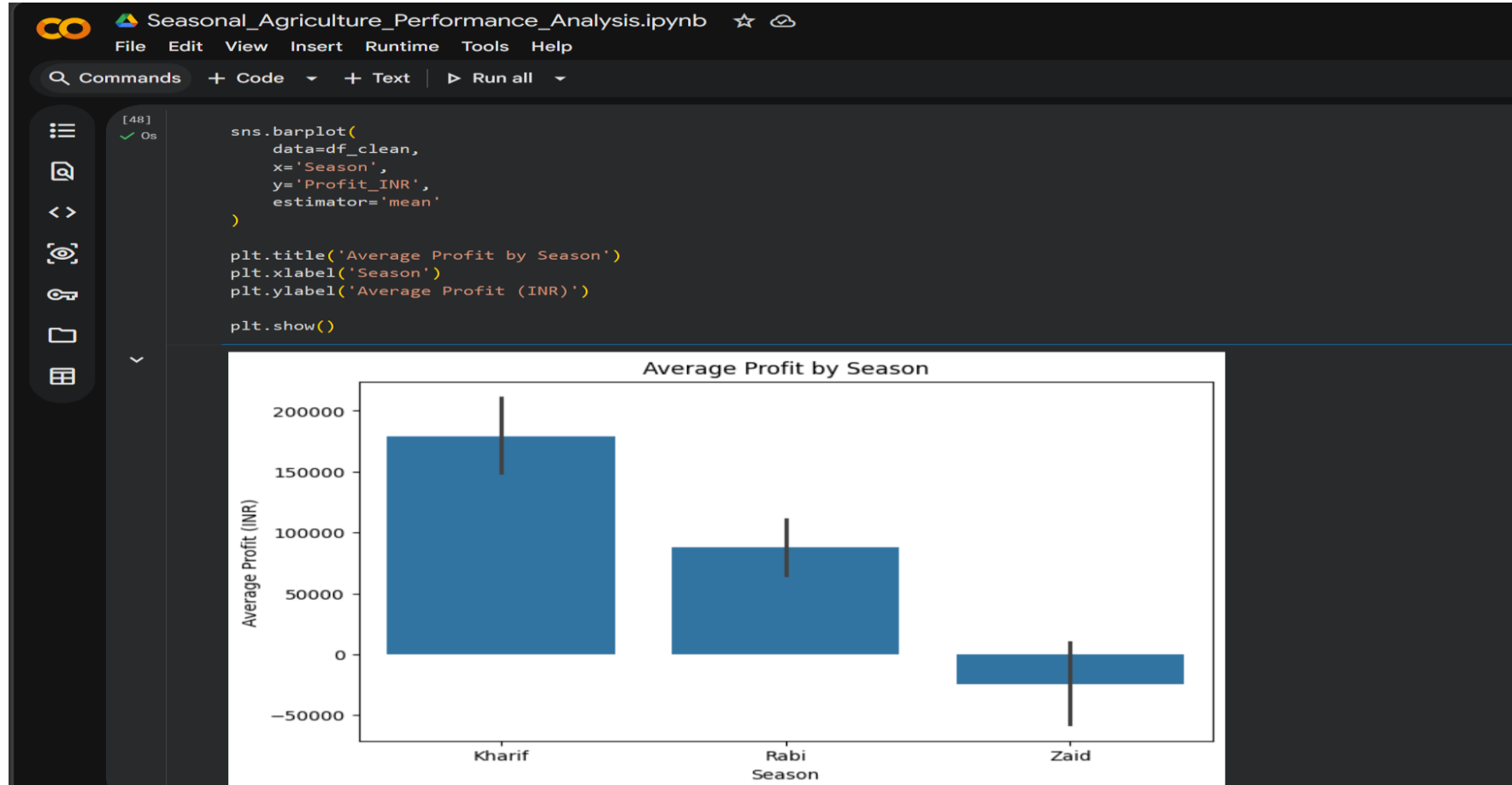
[6] df.shape
    (4000, 28)

[7] print("Number of Rows:", df.shape[0])
    print("Number of Columns:", df.shape[1])
    Number of Rows: 4000
    Number of Columns: 28
```

The notebook interface includes a top bar with the file name 'Seasonal_Agriculture_Performance_Analysis.ipynb', a menu (File, Edit, View, Insert, Runtime, Tools, Help), and a toolbar with icons for commands, code, text, and running all cells. A sidebar on the left contains icons for file explorer, search, and other notebook functions. The right side of the interface shows RAM and Disk usage indicators.

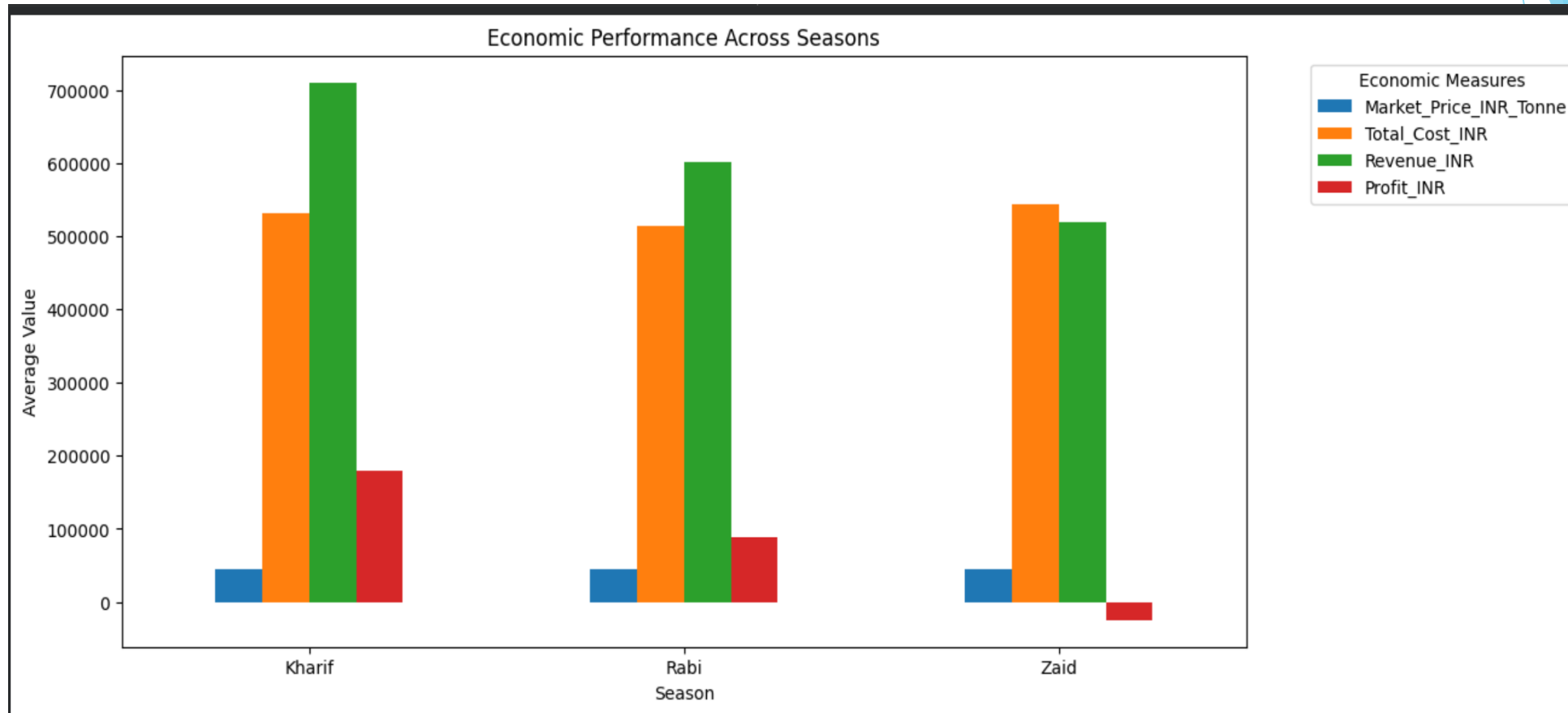
[The dataset contains 4,000 farm records and 28 agricultural attributes. The analysis includes crop, season, rainfall, temperature, soil conditions, irrigation, yield, production, cost, revenue and profit.]

RESULTS : Seasonal Performance



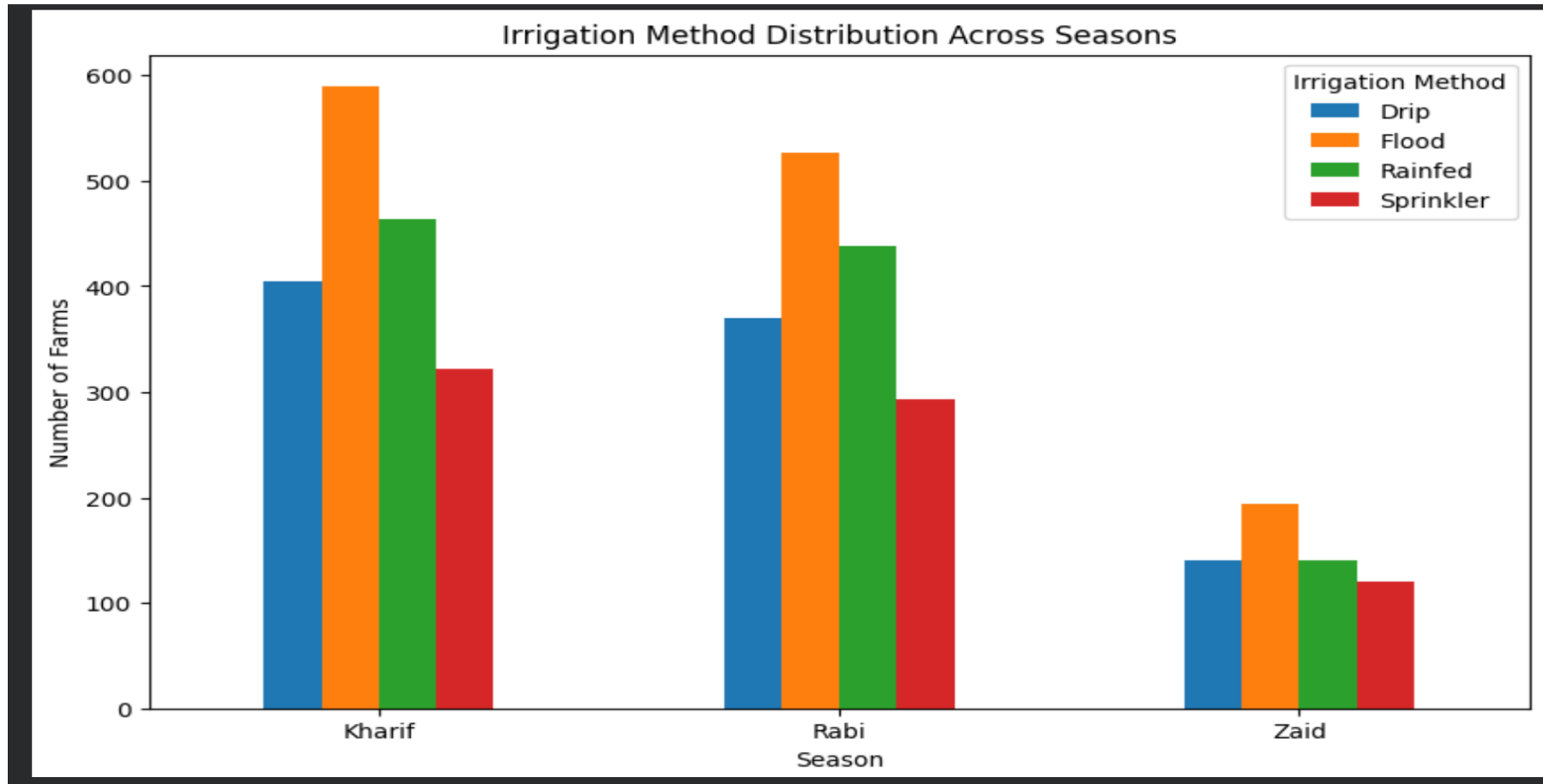
[Seasonal analysis was performed to compare agricultural profitability across Kharif, Rabi and Zaid seasons. The comparison helps identify 9/5/2026 Annual Review 7 which season provides better overall farm performance.]

RESULTS :Economic Performance Across Seasons



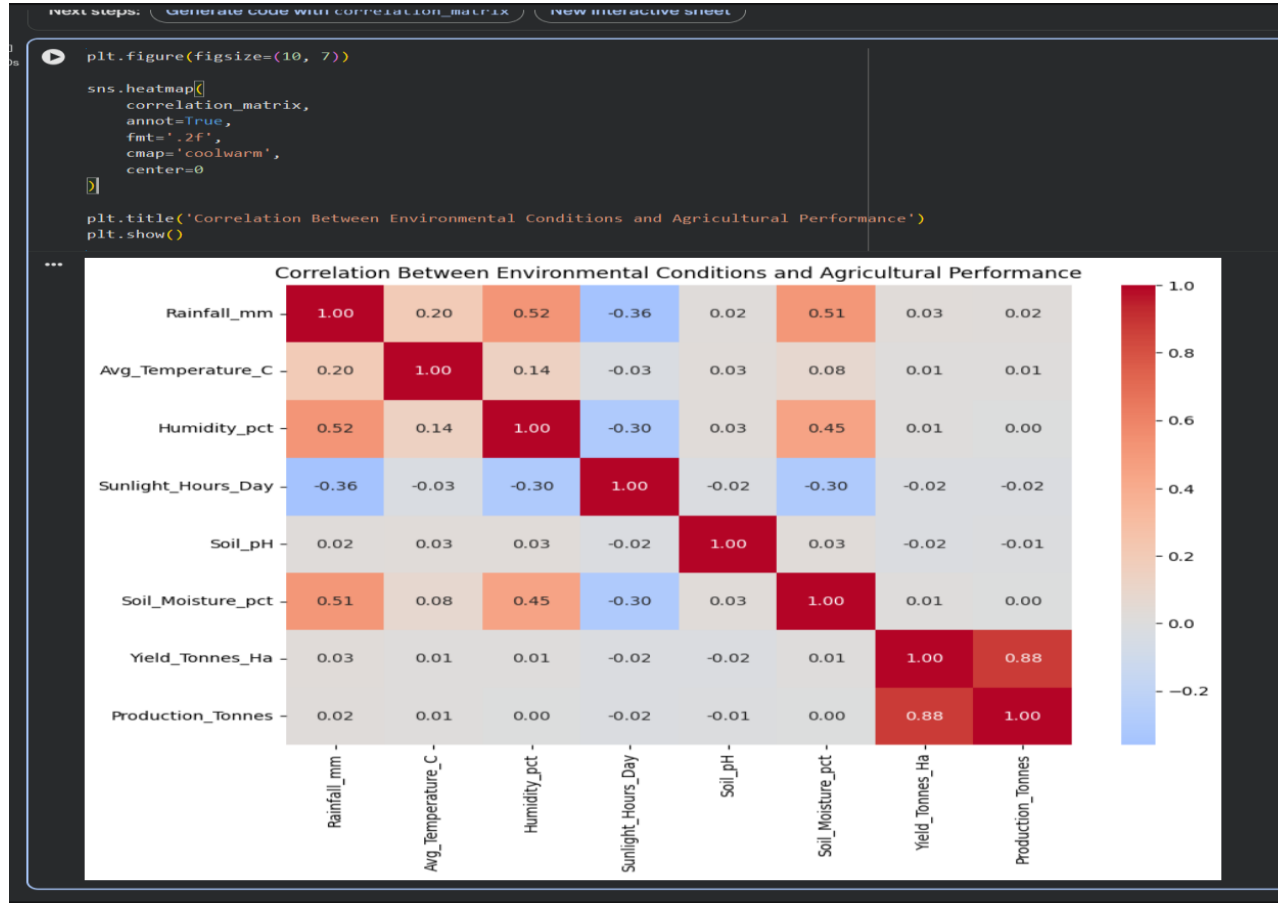
[Economic outcomes vary significantly across seasons. Kharif provides the highest average profitability, Rabi shows moderate profitability, while Zaid shows an average loss. Seasonal planning should therefore consider both expected revenue and production costs rather than market price alone.]

RESULTS : Irrigation Method Distribution Across Seasons



{The data shows that agricultural activities are not distributed equally across seasons. Crop participation and irrigation practices differ between seasons, with Rice being consistently prominent and Flood irrigation being the most commonly observed irrigation method.}

RESULTS : Correlation Analysis



[Correlation analysis was performed on the numerical agricultural variables to identify relationships between rainfall, temperature, soil parameters, yield, production, profit and 9/5/2026 Annual Review water usage. The heatmap provides a quick overview of strong and weak relationships within 10 the dataset.]

Future scope

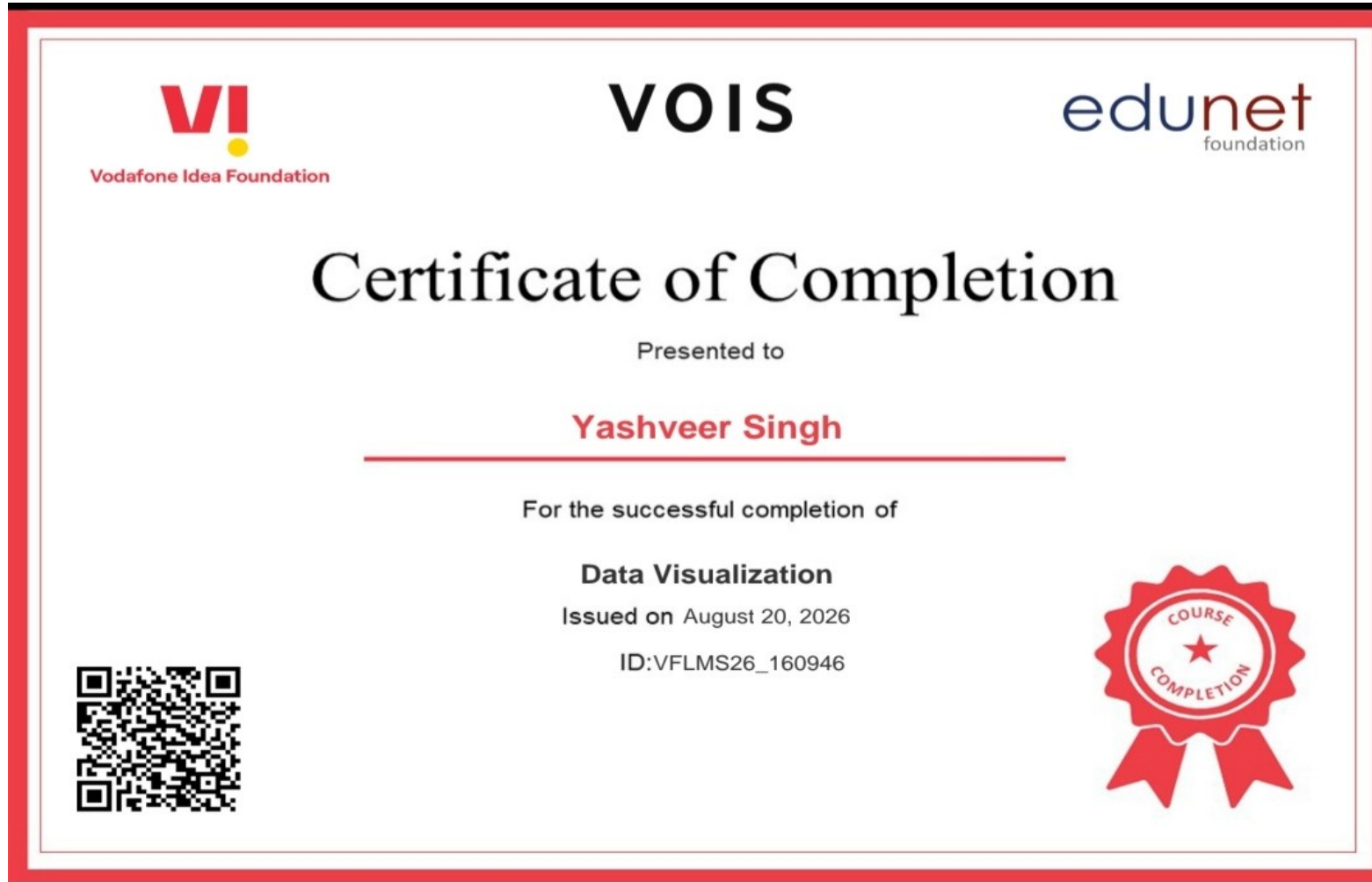
- Build an interactive Power BI dashboard.
- Add machine-learning based yield prediction.
- Develop crop recommendation systems.
- Predict agricultural profit before cultivation.
- Add weather forecasting data.
- Provide irrigation and resource optimization recommendations

GitHub Link

<https://github.com/Yashveer005/Seasonal-Agriculture-Performance-Analysis>

GitHub - [click here](#)

VOIS Course completion certificate - Course name- Data Visualization



Thank you

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VOIS for Tech / AICTE Major Project

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